

A Subgroup Obstruction Calculus for Two-Generator Exponential Congruences

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Abstract

We develop an elementary finite-group calculus for local congruence obstructions in exponential prime problems of the form

$$ma^kb^\ell + 1.$$

The main point is that the local obstruction at a modulus d is exactly a subgroup-membership obstruction for

$$\Gamma_d(a, b) = \langle a, b \rangle \subset (\mathbb{Z}/d\mathbb{Z})^*.$$

This gives exact local densities, exact mean-zero and second-moment identities, and a finite Fourier expansion through the character annihilator of $\Gamma_d(a, b)$. The motivating example is the Erdős–Graham Problem #203 sequence

$$m2^k3^\ell + 1.$$

All results in this note are elementary finite algebra and probability. No prime-distribution theorem and no claim about the full Erdős–Graham problem is used.

1 Introduction

Let $a, b \in \mathbb{Q}^*$, and consider congruences of the form

$$ma^kb^\ell + 1 \equiv 0 \pmod{d}.$$

When a, b, m are units modulo d , this congruence is equivalent to

$$a^kb^\ell \equiv -m^{-1} \pmod{d}.$$

Thus local solvability is controlled by whether $-m^{-1}$ lies in the subgroup generated by a and b .

For Erdős–Graham Problem #203, $a = 2$, $b = 3$, and $(m, 6) = 1$. The local group is

$$\Gamma_d = \langle 2, 3 \rangle \subset (\mathbb{Z}/d\mathbb{Z})^*.$$

The exact local obstruction density is

$$\rho_m(d) = \frac{\mathbf{1}_{\Gamma_d}(-m^{-1})}{|\Gamma_d|}.$$

The purpose of this note is to record the finite algebra behind this formula and its exact moments.

2 Main theorem package

The paper proves four finite statements.

1. A finite abelian subgroup obstruction lemma: normalized subgroup indicators have exact mean, variance, and character expansion.
2. A multiplication-fiber theorem: the fiber size of $H_1 \times H_2 \rightarrow H_1 H_2$ is exactly $|H_1 \cap H_2|$.
3. An exact local density theorem for $ma^k b^\ell + 1$ modulo d .
4. Exact moment and weighted-variance identities over the parameter m .

No analytic number theory is needed for these statements.

3 Finite abelian subgroup obstruction calculus

Definition 1. Let G be a finite abelian group and $\Gamma \leq G$. Define

$$\Delta_\Gamma(c) = \frac{\mathbf{1}_\Gamma(c)}{|\Gamma|} - \frac{1}{|G|}, \quad c \in G.$$

We call Δ_Γ the centered subgroup obstruction attached to Γ .

Theorem 1 (Finite abelian subgroup obstruction lemma). For a finite abelian group G and subgroup $\Gamma \leq G$,

$$\sum_{c \in G} \Delta_\Gamma(c) = 0,$$

and

$$\sum_{c \in G} |\Delta_\Gamma(c)|^2 = \frac{1}{|\Gamma|} - \frac{1}{|G|}.$$

Moreover, if $\widehat{G}$ is the character group and

$$\Gamma^\perp = \{\chi \in \widehat{G} : \chi(\gamma) = 1 \ \forall \gamma \in \Gamma\},$$

then

$$\Delta_\Gamma(c) = \frac{1}{|G|} \sum_{\substack{\chi \in \Gamma^\perp \\ \chi \neq 1}} \chi(c).$$

Proof. The mean-zero identity follows from

$$\sum_{c \in G} \frac{\mathbf{1}_\Gamma(c)}{|\Gamma|} = 1 \quad \text{and} \quad \sum_{c \in G} \frac{1}{|G|} = 1.$$

For the second moment,

$$\sum_{c \in G} \left(\frac{\mathbf{1}_\Gamma(c)}{|\Gamma|} - \frac{1}{|G|} \right)^2 = |\Gamma| \left(\frac{1}{|\Gamma|} - \frac{1}{|G|} \right)^2 + (|G| - |\Gamma|) \frac{1}{|G|^2},$$

which simplifies to $1/|\Gamma| - 1/|G|$.

For the character expansion, finite Fourier inversion gives

$$\frac{\mathbf{1}_\Gamma(c)}{|\Gamma|} = \frac{1}{|G|} \sum_{\chi \in \Gamma^\perp} \chi(c).$$

Subtracting $1/|G|$, the contribution of the trivial character, gives the claimed formula. $\square$

Definition 2 (Annihilator rank and obstruction energy). *Define the annihilator index*

$$\kappa_\Gamma = \frac{|G|}{|\Gamma|} = |\Gamma^\perp|$$

and the obstruction energy

$$\mathcal{E}(\Gamma) = \frac{1}{|\Gamma|} - \frac{1}{|G|} = \frac{\kappa_\Gamma - 1}{|G|}.$$

Corollary 1 (Energy monotonicity). *If $\Gamma_1 \leq \Gamma_2 \leq G$, then*

$$\mathcal{E}(\Gamma_2) \leq \mathcal{E}(\Gamma_1).$$

Proof. This is immediate from $|\Gamma_1| \leq |\Gamma_2|$. $\square$

4 Multiplication fibers

Lemma 1 (Multiplication fiber theorem). *Let G be an abelian group, and let $H_1, H_2 \leq G$. For $c \in G$,*

$$|\{(x, y) \in H_1 \times H_2 : xy = c\}| = |H_1 \cap H_2| \mathbf{1}_{H_1 H_2}(c).$$

Proof. The multiplication map

$$\mu : H_1 \times H_2 \rightarrow H_1 H_2, \quad (x, y) \mapsto xy$$

is surjective. Its kernel is

$$\{(u, u^{-1}) : u \in H_1 \cap H_2\},$$

which has size $|H_1 \cap H_2|$. Hence every element of $H_1 H_2$ has exactly $|H_1 \cap H_2|$ preimages, and elements outside $H_1 H_2$ have none. $\square$

5 Two-generator exponential congruences

Let $d \geq 1$, and suppose $a, b, m \in (\mathbb{Z}/d\mathbb{Z})^*$. Define

$$G_d = (\mathbb{Z}/d\mathbb{Z})^*, \quad \Gamma_d(a, b) = \langle a, b \rangle \subset G_d.$$

Proposition 1 (Exact local density). *The proportion of exponent pairs in the finite orbit $\langle a \rangle \times \langle b \rangle$ satisfying*

$$ma^k b^\ell + 1 \equiv 0 \pmod{d}$$

is

$$\rho_{m,a,b}(d) = \frac{\mathbf{1}_{\Gamma_d(a,b)}(-m^{-1})}{|\Gamma_d(a,b)|}.$$

Proof. The congruence is equivalent to $a^k b^\ell \equiv -m^{-1} \pmod{d}$. Apply Lemma 1 to

$$H_1 = \langle a \rangle, \quad H_2 = \langle b \rangle.$$

The number of pairs mapping to a target c is $|H_1 \cap H_2| \mathbf{1}_{\Gamma_d(a,b)}(c)$. Dividing by $|H_1||H_2|$ and using

$$|\Gamma_d(a,b)| = \frac{|H_1||H_2|}{|H_1 \cap H_2|}$$

gives the formula. □

6 Moments over m

For $m \in G_d$, define

$$\Delta_m(d; a, b) = \frac{\mathbf{1}_{\Gamma_d(a,b)}(-m^{-1})}{|\Gamma_d(a,b)|} - \frac{1}{\varphi(d)}.$$

Theorem 2 (Exact moment laws). *For every d with $a, b \in G_d$,*

$$\sum_{m \in G_d} \Delta_m(d; a, b) = 0,$$

and

$$\sum_{m \in G_d} |\Delta_m(d; a, b)|^2 = \frac{1}{|\Gamma_d(a,b)|} - \frac{1}{\varphi(d)}.$$

Proof. The map $m \mapsto -m^{-1}$ is a bijection of G_d . The result is therefore Theorem 1 with $G = G_d$ and $\Gamma = \Gamma_d(a,b)$. □

7 Character expansion

Theorem 3 (Character-annihilator expansion). *For $G_d = (\mathbb{Z}/d\mathbb{Z})^*$ and $\Gamma_d(a,b) = \langle a, b \rangle$,*

$$\Delta_m(d; a, b) = \frac{1}{\varphi(d)} \sum_{\substack{\chi \bmod d \\ \chi \neq \chi_0 \\ \chi(a)=\chi(b)=1}} \chi(-m^{-1}).$$

Proof. Characters $\chi \bmod d$ are the character group of G_d . The annihilator of $\Gamma_d(a,b)$ is exactly the set of characters satisfying $\chi(a) = \chi(b) = 1$. Apply Theorem 1. □

8 CRT orthogonality and weighted variance

Lemma 2 (CRT orthogonality). *Let $d = d_1 d_2$ with $(d_1, d_2) = 1$. If F_i has mean zero on $(\mathbb{Z}/d_i\mathbb{Z})^*$, then*

$$\mathbb{E}_{m \in (\mathbb{Z}/d\mathbb{Z})^*} F_1(m \bmod d_1) F_2(m \bmod d_2) = 0.$$

Proof. By the Chinese remainder theorem,

$$(\mathbb{Z}/d\mathbb{Z})^* \simeq (\mathbb{Z}/d_1\mathbb{Z})^* \times (\mathbb{Z}/d_2\mathbb{Z})^*,$$

and the expectation factors into the product of the two means. □

Theorem 4 (Prime-level weighted variance). *Let $\mathcal{Q}$ be a finite set of primes not dividing ab , and put $P = \prod_{q \in \mathcal{Q}} q$. Then*

$$\mathbb{E}_{m \in (\mathbb{Z}/P\mathbb{Z})^*} \left| \sum_{q \in \mathcal{Q}} \lambda_q \Delta_m(q; a, b) \right|^2 = \sum_{q \in \mathcal{Q}} |\lambda_q|^2 \frac{1}{q-1} \left(\frac{1}{|\Gamma_q(a, b)|} - \frac{1}{q-1} \right).$$

Proof. Expand the square. The cross terms vanish by Lemma 2 and the mean-zero identity. The diagonal terms are the normalized versions of Theorem 2 at prime modulus q . $\square$

9 Specialization to $a = 2, b = 3$

For $q \nmid 6$, write

$$\Gamma_q = \langle 2, 3 \rangle \subset \mathbb{F}_q^*, \quad a_q = \text{ord}_q(2), \quad b_q = \text{ord}_q(3).$$

Since $\mathbb{F}_q^*$ is cyclic,

$$|\Gamma_q| = \text{lcm}(a_q, b_q).$$

For $c \in \mathbb{F}_q^*$,

$$|\{(x, y) \in \langle 2 \rangle \times \langle 3 \rangle : xy = c\}| = \gcd(a_q, b_q) \mathbf{1}_{\Gamma_q}(c).$$

Thus for $(m, 6q) = 1$,

$$\rho_m(q) = \frac{\mathbf{1}_{\Gamma_q}(-m^{-1})}{|\Gamma_q|},$$

and

$$\sum_{m \in \mathbb{F}_q^*} \left| \rho_m(q) - \frac{1}{q-1} \right|^2 = \frac{1}{|\Gamma_q|} - \frac{1}{q-1}.$$

10 RMS obstruction law

At prime modulus $q \nmid ab$, Theorem 2 gives the normalized RMS size

$$\text{RMS}_m(\Delta_m(q; a, b)) = \left[\frac{1}{q-1} \left(\frac{1}{|\Gamma_q(a, b)|} - \frac{1}{q-1} \right) \right]^{1/2}.$$

Thus the local obstruction is governed exactly by the subgroup index

$$[(\mathbb{F}_q^*) : \Gamma_q(a, b)].$$

If $|\Gamma_q(a, b)| \asymp q$, then the RMS size is $O(q^{-1})$. If the subgroup is smaller, the RMS size is amplified by

$$\left(\frac{q-1}{|\Gamma_q(a, b)|} \right)^{1/2}.$$

11 Multi-generator extension

The same formalism applies to t -generator exponential congruences. Let

$$a_1, \dots, a_t \in (\mathbb{Z}/d\mathbb{Z})^*$$

and put

$$\Gamma_d(a_1, \dots, a_t) = \langle a_1, \dots, a_t \rangle \subset (\mathbb{Z}/d\mathbb{Z})^*.$$

For congruences

$$ma_1^{n_1} \cdots a_t^{n_t} + 1 \equiv 0 \pmod{d},$$

the exact local density is

$$\rho_{m,a_1,\dots,a_t}(d) = \frac{\mathbf{1}_{\Gamma_d(a_1,\dots,a_t)}(-m^{-1})}{|\Gamma_d(a_1,\dots,a_t)|},$$

and all moment and character-expansion identities above remain valid with $\Gamma_d(a, b)$ replaced by $\Gamma_d(a_1, \dots, a_t)$.

12 Concrete prime examples

These examples are finite checks and are included to make the obstruction calculus tangible.

12.1 No obstruction when $\Gamma_q = \mathbb{F}_q^*$

For $q = 5$,

$$\mathbb{F}_5^* = \{1, 2, 3, 4\}, \quad \langle 2, 3 \rangle = \mathbb{F}_5^*.$$

Thus

$$\rho_m(5) = \frac{1}{4}$$

for every $m \in \mathbb{F}_5^*$, and

$$\Delta_m(5) = 0.$$

The moment identity gives

$$\frac{1}{|\Gamma_5|} - \frac{1}{5-1} = \frac{1}{4} - \frac{1}{4} = 0.$$

12.2 A proper subgroup obstruction

For $q = 23$, one has

$$|\langle 2, 3 \rangle \bmod 23| = 11.$$

Thus the local density is

$$\rho_m(23) = \begin{cases} 1/11, & -m^{-1} \in \langle 2, 3 \rangle, \\ 0, & -m^{-1} \notin \langle 2, 3 \rangle. \end{cases}$$

The second moment is exactly

$$\frac{1}{11} - \frac{1}{22} = \frac{1}{22}.$$

This example shows the obstruction energy directly: the subgroup has index 2, and the centered variance records that index.

12.3 Interpretation

The obstruction is not mysterious. It is exactly the failure of the target residue $-m^{-1}$ to lie in the subgroup generated by the allowed exponential bases.

13 Independent verification of examples

The examples above are verified by the included script `verify_finite_examples.py`. In particular:

$$\langle 2, 3 \rangle \bmod 5 = \mathbb{F}_5^*,$$

and

$$|\langle 2, 3 \rangle \bmod 23| = 11, \quad \text{ord}_{23}(2) = 11, \quad \text{ord}_{23}(3) = 11.$$

Thus the $q = 23$ example has obstruction energy $1/22$, exactly as stated.

14 Further directions

The calculus above is elementary, but it supplies a reusable foundation for density-zero and conditional Kummer-distribution results in problems involving $ma^{k_\ell}b^\ell + 1$. The analytic problem of proving strong enough prime distribution in Kummer families is deliberately not claimed here.

References

- [1] H. Iwaniec and E. Kowalski. *Analytic Number Theory*. American Mathematical Society Colloquium Publications, 2004.
- [2] F. Pappalardi. On the order of finitely generated subgroups of $\mathbb{Q}^*$ modulo p and divisors of $p - 1$. *Journal of Number Theory*, 57(2):207–222, 1996.